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Student Teachers' Practice Self-Efficacy Prior to Their First Field Practice in Schools: Interrelatedness of Subconstructs Within Three Domains of Practice

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Abstract

Developing a strong sense of self-efficacy is thought to be decisive for student teachers to help them prepare for challenges in the profession. However, while most recognize that confidence in teaching involves multiple and interrelated dimensions, we know little about the interplay between dimensions, hindering our understanding of how self-efficacy is nurtured and affected. This study examines how different self-efficacy dimensions relate, using a recently developed and targeted measure of Practice Self-Efficacy (PSE) for student teachers. With chain graph models, used on survey data from 405 first-year student teachers from a Danish University College, we find that most, but not all, PSE dimensions are related, with the strongest associations found among dimensions within the Teaching-related domain. Most PSE dimensions were not associated with student teachers' background characteristics, except for *Differentiation PSE* and *Teaching in itself PSE*. Thus, teaching experience prior to teacher education was negatively associated with *Differentiation PSE*, whereas it was positively associated with *Teaching in itself PSE* and with differing strengths dependent on chosen teaching major. The results point to possible areas in teacher education where interventions may effectively enhance practice self-efficacy among first-year students.

Keywords: practice self-efficacy; teacher education; field practice; chain graph model

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1. Introduction

In Bandura's social cognitive theory, self-efficacy refers to individuals' beliefs in their own capabilities to plan and perform actions necessary to attain a certain outcome (Bandura, 1997). Self-efficacy affects how individuals feel, think, and behave, but also feelings, cognitive processes and behavior can affect self-efficacy, as the relationships are reciprocal (Bandura, 1997). Thus, low self-efficacy, which is in itself a feeling of insecurity, has been found to be associated with an increased risk of depression and anxiety (Muris, 2002; Tahmassian & Jalali Moghadam, 2011). High self-efficacy has been found to facilitate cognitive processes, such as decision making and learning (Maddux, 2002; Scholz et al., 2002; Schunk & DiBenedetto, 2020). With respect to behavior, the self-efficacy level has been found to be part of motivation processes and in determining how actions are planned and carried out, and how persistent one is (Bandura, 1997; Tinto, 2017; Wentzel & Miele, 2009).

In relation to the fields of education and teaching, self-efficacy has been construed in varying contexts and somewhat role-specific forms: Teacher Self-Efficacy (TSE), as a form

of workplace self-efficacy influencing occupational decision making, career choice, career development, and career trajectory (Bandura, 1997; Klassen & Chiu, 2011). Academic self-efficacy (ASE), as a form of student learning self-efficacy, which has been found to influence educational outcomes, persistence and intention to drop out or stay (Bager-Elsborg et al., 2019; Khine & Nielsen, 2022; Richardson et al., 2012). Practice self-efficacy (PSE), as a form of student learning self-efficacy complimentary to ASE, is concerned with self-efficacy in relation to the learning of the practical and performative skills and competences necessary in professions, such as, for example, the teaching profession (Nielsen et al., 2024a, 2024b).

Developing a strong sense of self-efficacy—whether ASE or PSE—is assumed to be critical for student teachers to help them adapt to the demands of the profession. Beyond the necessary pedagogical skills, self-efficacy is considered essential because it is expected to strengthen courage and motivation to implement high-quality teaching and persistence when things get tough (Tschannen-Moran & Hoy, 2001). Research supports this idea and finds that teachers with high self-efficacy are linked to better student academic achievement (K. R. Kim & Seo, 2018; Caprara et al., 2006), likely because they implement effective teaching strategies and foster stronger student motivation (Klassen & Tze, 2014; Ashton & Webb, 1986; Tschannen-Moran & Hoy, 2001). Moreover, evidence shows that self-efficacy not only serves as a shield against the early career 'reality shock' encountered in classrooms (H. Kim & Cho, 2014), but also increases teachers' resilience towards stress (Klassen & Chiu, 2011) and burnout (Betoret, 2006; Skaalvik & Skaalvik, 2007), thereby reducing the risk of an untimely career exit (Klassen & Chiu, 2011). In addition, the reciprocal causation proposed by Bandura (1997) means that self-efficacy (TSE, ASE or PSE) not only affects the emotional experiences of teachers and students. Their cognitive processes, and their behaviors, TSE, ASE and PSE can also be affected by emotional cognitive and behavioral experiences within the profession or the profession education.

While nurturing self-efficacy in teacher students is vital, it is key to recognize that confidence in teaching not only involves one, but multiple interrelated domains. Instead of reflecting a single belief, self-efficacy in teaching may span practices related to various aspects of teaching, such as classroom management, instructional strategies, student engagement, planning or technology integration, each with unique tasks and separate challenges (Tschannen-Moran & Hoy, 2001; Rupp & Becker, 2021; Elstad & Christophersen, 2017). Bandura (1997) originally emphasized this multidimensional view, thereby alluding to the idea that dimensions may reinforce one another, while others operate independently, i.e., that confidence in one dimension does not influence or transfer to another. Recognizing the possible interplay between dimensions appears crucial to understand how self-efficacy can be affected and nurtured.

However, still relatively little is known about how different self-efficacy dimensions interact—especially among student teachers. Previous research has not neglected this topic (e.g., Klassen & Chiu, 2011; Berg et al., 2023), but rather faced two important methodological limitations. First, many studies have relied on self-efficacy measures not designed for student teachers, overlooking the important context in which they develop and are able to assess their confidence—through field practice—thereby risking capturing something very different, e.g., a generalized personality trait rather than relevant context-specific self-efficacy beliefs (Pajares, 1996). Secondly, most previous studies have examined simple statistical associations, making us unable to disentangle multivariate relationships and root out confounding factors. As a result, we do not have a clear picture of how self-efficacy dimensions interact and affect each other in the early stages of becoming a teacher.

1.1. Relationships Between Self-Efficacy Dimensions Among Student Teachers

Research on the relationship between teacher efficacy dimensions has utilized different self-efficacy measures, likely a result of the various ways it has been defined (Duffin et al., 2012).

One of the most widely used measures, the Teachers' Sense of Efficacy Scale (TSES), was developed by Tschannen-Moran and Hoy (2001) to measure teachers perceived capabilities in key domains of teaching. The TSES consists of three dimensions in teaching: (1) classroom management (e.g., *"How much can you do to control disruptive behavior in the classroom?"*), (2) instructional strategies (e.g., *"To what extent can you use a variety of assessment strategies?"*), and (3) student engagement (e.g., *"How much can you do to get students to believe they can do well in schoolwork?"*) (Tschannen-Moran & Hoy, 2001, p. 800). The measure was initially developed for and with practicing teachers (Tschannen-Moran & Hoy, 2001, p. 795), but it has been used among both practicing and student teachers.

Tschannen-Moran and Hoy (2001) initially examined the interrelationships among the above three dimensions using a pooled sample of practicing (n = 255) and student teachers (n = 155), finding that the three dimensions were all positively correlated. Using the 12-item short form, their findings showed moderate, positive correlations, assumed to be Pearson's r as it is unspecified (Unless explicitly stated otherwise, all reported correlation coefficients in this subsection are Pearson's r. This is, however, an assumption, because the reviewed articles typically do not specify the type of correlation). r = 0.46 (instructional strategies and classroom management), r = 0.61 (instructional strategies and student engagement), and r = 0.50 (classroom management and student engagement). Among practicing teachers, research consistently finds similar relationships (e.g., Heneman et al., 2006; Klassen et al., 2009; Klassen & Chiu, 2011). Among student teachers, however, the evidence is scarce. Although debate exists about the usefulness of the three-dimensional measure for student teachers (Duffin et al., 2012)—a point we revisit later—findings by Klassen and Chiu (2011) from Canada suggest slightly stronger relationships among student teachers (r = 0.52–0.62), while Poulou (2007) reports comparable results among student teachers in Greece.

Few studies examine the relationship between teacher self-efficacy dimensions among student teachers using either slightly modified versions of the TSES (Burgueño et al., 2019; Khairani & Makara, 2020) or more freely derived variations thereof. For instance, Burgueño et al. (2019) found a strong relationship between all three self-efficacy dimensions using a modified version of the short TSES-11 instrument in a sample of Spanish student teachers. This corresponds to the previously reported results among student teachers using the original TSES version. Using an even more flexible approach in a study with Swiss student teachers, Rupp and Becker (2021) adapted the instructional strategies subscale to measure self-efficacy for learning support and derived another dimension, lesson planning. Their results similarly showed a strong correlation between these two dimensions, which suggests that confidence in supporting student learning is closely related to confidence in lesson planning.

Another measure, the Norwegian Teacher Sense of Efficacy Scale (NTSES) (Skaalvik & Skaalvik, 2007), was originally developed in Norway with practicing teachers, but has also been translated and used in other countries (e.g., Avanzi et al., 2013; Berg et al., 2023). The NTSES captures six key dimensions of teachers' work: Instruction, Adapting Education to Individual Students' Needs, Motivating Students, Keeping Discipline, Cooperating with Colleagues and Parents, and Coping with Changes and Challenges. Mostly applied among practicing teachers, studies consistently find positive correlations among all six dimensions of teacher self-efficacy. In Norway, Skaalvik and Skaalvik (2007) reported zero order correlations of moderate strength between dimensions (r = 0.33–0.54), whereas Avanzi et al. (2013) also found positive correlations among all six subscales in Italy. Notably, a

single study among student teachers suggests even stronger associations between domains in Norway (r = 0.60–0.79) and in New Zealand (r = 0.71–0.88) (Berg et al., 2023).

According to the above body of work, the message seems clear: if teachers, and, in particular, student teachers, are confident in one area of teaching, they tend to be confident in others as well—across measures and contexts. Importantly, however, this research has not considered an important constraint: student teachers are not fully-fledged teachers, yet existing measures are designed for this group and may not be valid for student teachers. This limitation is evident in Duffin et al. (2012) who find no support of the three-dimensional structure of the TSES among student teacher (Tschannen-Moran and Hoy (2001) also originally hinted at such problems with the TSES.). Extending this critique, Nielsen et al. (2024a) argue that student teachers' self-efficacy should be measured within the context where they apply their teaching practices and their self-efficacy develops—field practice that is—with a focus on the unique tasks and challenges they face. Without a context-specific measure tailored to student teachers' actual experiences, measurements risk being inaccurate or reflecting more general personality traits rather than self-efficacy (Pajares, 1996). In response, Nielsen et al. (2024b) developed and validated the Practice Self-Efficacy Questionnaire (PSEQ), a targeted instrument to measure student teachers' self-efficacy in the context of field practice, consisting of seven dimensions: Teaching-related PSE with the subscales *Planning and preparation*, *Teaching in itself*, *Class management*, and *Differentiation*; Relational PSE with the subscales *Pupils* and *Adult collaborators* (parents and co-teachers); and Evaluative and developmental PSE with the single subscale *Evaluation and development*. Critical to the aim of this study, however, is that neither this instrument nor any other instrument tailored for student teachers have—to our knowledge—been used to examine the relationship between the domains of student teachers' self-efficacy.

Another limitation in prior research on student teachers is that few account for confounders when examining relationships between self-efficacy across domains. This makes it hard to disentangle whether observed relationships are real or spurious. For example, an observed positive association between self-efficacy in classroom management and instruction may be genuine, but it could also be driven by background characteristics, such as prior teaching experience. If experience enhances both self-efficacy domains and is not adjusted for, the observed association may be overestimated or completely absent. Similarly, self-efficacy domains may act as confounders themselves, creating apparent relationships between other domains. For instance, confidence in classroom management could inflate the relationship between self-efficacy in instruction and differentiation, if self-efficacy in classroom management spreads to one's perceived ability in the other domains. Importantly, some studies do address this concern by using latent correlations from Confirmatory Factor Analysis (CFA) (Poulou, 2007; Burgueño et al., 2019; Khairani & Makara, 2020), which account for the influence of other self-efficacy domains. Yet, these studies do not control for background factors, thereby neglecting possible confounders such as differences in student characteristics.

1.2. Student Characteristics with Possible Effect on Student Teachers' Practice Self-Efficacy

1.2.1. Previous Teaching Experience

Previous research on the criterion validity of the seven individual PSEQ subscales has shown teaching experience prior to teacher education to be associated with significantly higher scores on all subscales than no prior teaching experiences (Nielsen et al., 2024a). Effect sizes were in most cases small, with some cases getting close to medium effects (0.20 < Cohen's d < 0.50), and only the *Teaching in itself* subscale having a larger than medium effect (Cohen's d = 0.62). Furthermore, research on motivations to enter teacher education and become a teacher—among prospective student teachers—has shown that the

main motivators are closely linked with self-efficacy. Thus, the intrinsic value of teaching, the social value of teaching, and the perception of own teaching abilities, which are all affected by and affect self-efficacy, have been found to be the three of four most important motives for choosing a teaching career (Pavin Ivanec, 2023; Takala et al., 2024).

1.2.2. Admission Track

While little is known about how admission track relates to student teachers' self-efficacy, a few studies have examined the relationships between admission track and academic self-efficacy in higher education more broadly. O'Neill and Nielsen (2024) found that Pre-Academic *Learning* Self-Efficacy (PAL-SE) measured at study start were higher for a sample of students from five different Bachelor programs admitted on other qualifications (in that case via a test-based admission track) compared to those admitted via the grade-based admission track. Makransky et al. (2017) found that mid-semester levels of Specific Academic *Learning* Self-Efficacy (SAL-SE) were higher for a sample of psychology students admitted on other qualifications (in that case via a test-based admission track) compared to those admitted via the grade-based admission track. Nielsen et al. (2017) found evidence of a more complex pattern of differences in SAL-SE and Specific Academic *Exam* Self-Efficacy (SAE-SE) among psychology students, depending on both admission track and university. At a university where the two admission tracks differ substantially, students admitted through the other qualifications track showed a larger difference between SAL-SE and SAE-SE scores compared to those admitted through the grade-based track. At a university where the two admissions tracks were more alike, there were no significant differences between SAL-SE and SAE-SE scores given admission track. We have not identified any studies on differences in practice self-efficacy for student teachers' dependent on admission track.

1.2.3. Major Teaching Subject

Few studies have investigated whether student teachers' self-efficacy differs based on major or teaching subject(s). A study with 379 student teachers from Canada, Klassen and Chiu (2011) found that student teachers who plan to teach, applied or more vocational subjects feel less efficacious in classroom management and instructional strategies compared to those with more traditional academic teaching subjects such as English, Mathematics, Science or Arts. In a more recent study, Klempin et al. (2019) examined if German student teachers' self-efficacy changed during a complexity-reduced teacher training format across four teaching subject domains: English, History, Physics, and Primary Education. Although the student teachers' self-efficacy remained stable across the training period, the study highlighted, importantly, that there were no substantial differences in self-efficacy based on academic subject—neither pre- nor post-intervention. Thus, although student teachers' self-efficacy may vary by major teaching subject, it seems to mainly differ between vocational and academic teaching subjects.

1.2.4. Gender

Research on gender differences in academic self-efficacy is prolific. In a meta-analysis of 247 independent studies, spanning primary school to university, students and teachers, and multiple countries (N = 68,429) Huang (2013) found effect sizes ranging from −1.60 to 1.40, with a very small mean effect size (Hedge's g = 0.08) favoring males. A more recent study of gender differences in in-service teachers' self-efficacy beliefs as they related to questions on ability to complete teaching responsibilities, such as planning lessons within the curriculum and managing their classroom effectively, found no gender differences in self-efficacy (Strunc & Murray, 2019). In a prediction study on student teachers' academic self-efficacy in the Danish teacher education context (N = 407), Nielsen (2022) found a medium strong correlation between gender and pre-academic learning self-efficacy,

measured just prior to their starting teacher education, indicating that male student teachers scored higher ($\gamma = -0.24$). Also in the Danish teacher education context, Nielsen et al. (2024b) found weak to moderate strong gamma correlations between gender and five subscales of the PSEQ, all favoring male student teachers. Specifically, weak correlations were observed in the Class Management ($\gamma = 0.18$) and Teaching in itself ($\gamma = 0.16$) subscales, whereas medium correlations were found in the Planning and preparation ($\gamma = 0.21$), Adult collaborators ($\gamma = 0.22$), and Pupils (Relational) ($\gamma = 0.23$) subscales.

1.3. The Current Study

1.3.1. Context

The study was conducted within the context of the Danish teacher education for primary and lower secondary school, which is a four-year degree program equivalent of 240 European Credit Transfer System (ECTS) points. Teacher education is in Denmark offered at University Colleges and leads to a Professional bachelor's degree (Act on the Education of Teachers, 2024). As in the other Nordic countries, the Danish teacher education is an integrated program focusing on the coherence between on-campus teaching and field practice (Weisdorf, 2020). In other words, the study program requires that teacher students acquire knowledge of academic, pedagogical and didactical subjects as well as the profession's core skills and practices. For the latter purpose, the Danish teacher education includes field practice in schools equivalent to 40 ECTS points. In comparison, teacher education in Sweden includes field practice equivalent to 30 ECTS, in Finland it is 20 ECTS, in Norway a minimum of 115 days (Weisdorf, 2020), and in Iceland there are no official regulations on field practice placements (Source: e-mail correspondence with the Icelandic Ministry of Education, Science and Culture). Field practice placements are included in each of the four years of the Danish degree program. During the first year, field practice involves a prolonged period of time with, for example, a succession of one-day school visits and/or short full time field practice placements (Act on the Education of Teachers, 2024).

The national learning objectives for first-year field practice are threefold: "*The student can identify and gather relevant knowledge on teachers' planning and teaching. The student can plan, implement and evaluate short teaching sequences, and is able to explain own didactical deliberations. The student can understand and reflect upon own role as a teacher*". Furthermore, they should acquire "*Beginning experiences with investigations and descriptions of own practice and the practice of others, beginning experiences with observation, beginning experiences with investigation of issues within the teaching profession.*" [author's translation; UCL Erhvervsakademi og Professionhøjskole, 2024, p. 6).

At the University College in this study, first-year field practice includes seven single-day school visits over 11 weeks, followed by three weeks of full-time practice, and ends with eight single-day visits over nine weeks, all at the same school (source: Internal documents provided by the field practice coordinator). An 80% accumulated attendance is required in the first year of the degree program. Field practice is mandatory, and the field practice is required to be approved by the school in order for the student teacher to be able to continue in the teacher education program (Act on the Education of Teachers, 2024).

1.3.2. Aims and Expectations

The aim of the current study was to conduct the first study of the interrelationships between the seven practice self-efficacy dimensions of the PSEQ (Nielsen et al., 2024b; Nielsen & Pettersson, 2025) for first-year student teachers prior to their field practice. The secondary aims were to test how many of the seven practice self-efficacy dimensions were associated with teaching experience acquired prior to starting teacher education, chosen

teaching major, students' basis for admittance to the teacher education, or their gender, when all seven PSE dimensions were considered simultaneously in a multivariate model.

Based on the theoretical framework of self-efficacy (Bandura, 1997) and previous research, we had both general and specific a priori expectations of relationships among practice self-efficacy dimensions and relationships between background variables and single practice self-efficacy dimensions.

Among the seven practice self-efficacy dimensions, we expected the following relationships, when the dimensions were all part of the same multivariate model.

Hypothesis 1: *We expected the Planning and preparation and Evaluation and development dimensions to be positively associated, because both constitutes forms of planning, and thus the same high-order cognitive abilities are required for this executive function, whether it is expressed in one way or the other (Cristofori et al., 2019). Thus, practice self-efficacy beliefs in these abilities within one dimension may reflect self-efficacy beliefs in the other as well (Bandura, 1997).*

Hypothesis 2: *We expected the Planning and preparation and Differentiation dimensions to be positively associated, because differentiation of teaching also entails planning, and thus, as above, both require the same high-order cognitive abilities (Cristofori et al., 2019). In other words, it would be difficult to be self-efficacious in relation to ability to differentiate teaching, if you do believe you are self-efficacious in regard to planning. Rupp and Becker (2021) showed results with another instrument to support this expectation.*

Hypothesis 3: *We expected the Class management, Teaching in itself and Pupils (relational) dimensions to be positively associated based on similar findings with other instruments with similar dimensions (e.g., Burgueño et al., 2019; Klassen & Chiu, 2011; Poulou, 2007). Furthermore, classroom management has previously been construed as the way in which an educator delivers the curriculum, lessons and the environment they provide for their pupils, and as a means of building effective relationships with pupils (Hans & Hans, 2017), and thus it was likely that these dimensions of practice self-efficacy would be positively associated.*

Furthermore, we expected some of the PSE subscales to depend on the included background variables in the following ways.

Hypothesis 4: *Based on research on gender differences in academic self-efficacy and practice self-efficacy for student teachers, we expected that some of the practice self-efficacy dimensions would be associated with gender, so that the male student teachers would show higher practice self-efficacy than the female student teachers, when all dimensions were included in the same model (Huang, 2013; Nielsen, 2022; Nielsen et al., 2024b; Strunc & Murray, 2019). More specifically, we expected this to be the case for the Planning and preparation, the Adult collaborators, and the Pupils (relational) dimensions, as these have been found to be most strongly associated with gender (Nielsen et al., 2024b).*

Hypothesis 5: *Based on research on the relationships between teaching experience prior to teacher education and the seven individual practice self-efficacy dimensions, we expected that some practice self-efficacy dimensions would be associated with prior teaching experience, when all dimensions were included in the same analysis (Nielsen et al., 2024b). In particular, we expected that the Teaching in itself dimension would be strongly associated with prior teaching experience as these experiences will likely have given the student teachers mastery experiences for different teaching practices, and, potentially even, vicarious experiences. It appears unlikely that very negative prior experiences would have resulted in them applying for teacher education as this would clash with known motivations to enter teacher education (Pavin Ivanec, 2023; Takala et al., 2024).*

Based on research on differences in academic self-efficacy, depending on admission track within the higher education context, we expected student teachers admitted on other qualifications to have higher practice self-efficacy than those admitted through the grade-based admission track (Makransky et al., 2017; Nielsen et al., 2017; O'Neill & Nielsen, 2024). However, as practice self-efficacy for student teachers is a new area within self-efficacy theory and research, we were not able to point to specific dimensions of practice self-efficacy, where we would expect this.

Based on previous research, we had no prior expectations concerning relationships between the Major teaching subject chosen by the student teachers and the seven practice self-efficacy dimensions.

2. Methods

2.1. Participants and Data Collections

Data were collected from first-year student teachers admitted to the teacher education program in one Danish University College in 2023 and 2024. Only students registered to commence their first-year field practice placement two weeks prior to its start were invited to participate (N = 586). Data were collected through online surveys utilizing student emails, as the teacher education advises students to check this email at least once a week. Students received an invitation email which described the purpose of the study, the relevant GDPR information such as students' rights to not participate, how to withdraw, etc. Students also received two reminders, and the surveys were closed when field practice started. The data collections resulted in a response rate of 69% (N = 405) (Table 1).

Table 1. Characteristics of the student sample (N = 405 [a]).

	Frequency (%)
Teacher education program	
Regular	383 (94.6)
Other [b]	21 (5.1)
Year of admittance	
2023	218 (53.8)
2024	187 (46.2)
Major teaching subject	
Danish	225 (55.6)
Mathematics	111 (27.4)
English	64 (15.8)
Previous completed education	
No	339 (83.7)
Yes	64 (15.8)
Previous teaching experience	
No	189 (46.7)
Yes	214 (52.8)
Context of teaching experience [c]	
Public school substitute teacher	167 (41.2)
High school	8 (2.0)
Academy Profession Degree Programs (APDs)	6 (1.5)
University college (e.g., teacher, nurse, and the like)	3 (0.7)
University	3 (0.7)
Single (public) lectures	27 (6.7)
Other [d]	73 (18.0)
Admission track	
Grade-based	160 (39.5)
Other qualifications	242 (59.8)

Table 1. *Cont.*

	Frequency (%)
Gender	
Female	252 (62.0)
Male	149 (36.8)
Other gender identification [e]	3 (0.7)
Mean age (SD), range	23.7 (5.1), 18–66

Notes. 202 of the students admitted in 2023 have previously been utilized as part of a sample with the purpose of validation of the seven subscales through for item analyses (Nielsen et al., 2024a). However, as the purpose of the current study is entirely different, the secondary use of this data is warranted. [a] for variables where categories do not add up to 405, it is due to a few missing responses on the variable in question. [b] includes STEM program, trainee program and distance (online) program. [c] context only provided for the 214 with teaching experience. However, it was possible to report more than one context. [d] additional pre-defined context not reported by any students was the Danish University Extension [Folkeuniversitet]. [e] excluded from analysis.

Among the participating student teachers, 53.8% were admitted to the degree program in 2023 and the rest in 2024 (Table 1). Almost 60% were admitted based on other qualifications than grade point average. Out of the three possible major subjects, consisting of Danish, Mathematics and English, more than half of the participants had chosen Danish (Major teaching subjects of 50 ECTS have to be chosen within three subjects; Danish, mathematics, and English. However, University colleges are not required to offer English as a major teaching subject). Slightly more than half of the participating students had teaching experience prior to being admitted to the teacher education, primarily in public schools (Danish state schools). A large majority had not previously completed higher education. The mean age of the participating students was 22.7 years, and 62% of the participating students were female.

2.2. Instrument

The Practice Self-Efficacy Questionnaire (PSEQ) for student teachers is a self-report instrument with 40 items measuring seven dimensions of practice self-efficacy within three domains of practice (Nielsen et al., 2024b). Domains of practice and subscales are as follows: Teaching-related PSE with the subscales *Planning and preparation* (7 items), *Teaching in itself* (7 items), *Class management* (6 items), and *Differentiation* (6 items); Relational PSE with the subscales *Pupils* (4 items) and *Adult collaborators* (parents and co-teachers) (4 items); and Evaluative and developmental PSE with the single subscale *Evaluation and development* (6 items). The original Danish PSEQ items are found in Nielsen and Pettersson (2025), while English—non-validated—translations of the items are provided in Nielsen et al. (2024b).

Students are asked *"In your upcoming field practice, how confident are you that you have sufficient abilities to . . ."*, and proceed to rate the item statements using a four-point response scale: very confident (3), confident (2), not particularly confident (1), not confident at all (0).

Nielsen et al. (2024b) found each of the seven subscales of the PSEQ to fit either the Rasch model (RM; Rasch, 1960; Christensen et al., 2013) or a graphical loglinear Rasch model (GLLRM; Kreiner & Christensen, 2007). Only the GLLRM for the *Class management* scale included an item (I4), which functioned differentially, while the Rasch or graphical loglinear Rasch models for the remaining scales were free from DIF. When a scale fits the RM, the unweighted sum score is a sufficient statistic for the estimated person parameters, a property only of Rasch models (Fischer, 1995). This sufficiency is retained in GLLRMs that are free of differential item functioning (Kreiner & Christensen, 2007). Thus, to ensure sufficiency of the sum scores, we chose to eliminate item 4 from the current study, utilizing 39 of the 40 PSEQ items in the analysis (see Section 2.3).

The score distributions of the seven scales are shown in Table 2. In relation to the spread of the data, the range statistics (i.e., the range, minimum, and maximum) show that

the full or nearly full score range was used by the student teachers across all subscales. The standard deviations (SD) confirm that all subscales provided considerable variation, with a coefficient of variation (CV; SD/M) ranging between 22 and 35%. The histograms show that the distributions are quite symmetric around the mean.

Table 2. Descriptive statistics for the seven practice self-efficacy subscales.

PSE Subscale (n)	Range	Min	Max	Median (IQR)	Mean	SD	Histogram
Planning and preparation (396)	0–21	1	21	14 (11–16)	13.3	3.68	
Teaching in itself (396)	0–21	5	21	15 (13–17)	14.7	3.24	
Differentiation (396)	0–18	5	18	12 (10–14)	11.9	2.71	
Class management (396)	0–15	3	15	10 (9–12)	10.4	2.29	
Pupils (396)	0–12	3	12	8 (7–10)	8.6	1.92	
Adult collaborators (399)	0–12	0	12	8 (6–9)	7.6	2.63	
Evaluation and development (396)	0–18	2	18	12 (10–14)	12.0	3.00	

Notes. IQR = Interquartile range. Complete cases across subscales (n = 396).

2.3. Analysis

For the analysis, we used so-called chain graph models which are log-linear graphical models (Lauritzen, 1996). The choice of these models over, e.g., structural equation models was made because the latter presumes at least interval level and normally distributed data. Log-linear chain graph models, on the other hand, are appropriate for counting and ordinal level scales and have no statistical requirements for specific distributions of the included variables (Gundelach & Kreiner, 2004), and as the PSE scales in the current study are ordinal (Nielsen et al., 2024b).

Chain graph models consist of nodes representing variables, directed edges (i.e., arrows) representing causal associations and undirected edges representing non-causal associations. Thus, the chain graph model is a generalization of the Directed Acyclic Graphs (DAGs; Digitale et al., 2022), as the latter do not include undirected edges (Lauritzen & Richardson, 2002). Chain graph models have a block-recursive structure, where arrows are present between blocks, and undirected edges are present within blocks. All paths between any two variables can be determined by the graph structure, thus making it possible to identify a minimum set of variables to condition on, when estimating the direct association between two variables, and thereby simplifying the analysis (for an introduction to graphical models see (Lauritzen, 1996) and (Kreiner et al., 2009)). Thus, with the chain graph models it was possible to assess not only the associations between the seven PSEQ subscales, as was our primary aim, but also, at the same time, test associations between the PSEQ subscales and background variables in a multivariate model with all subscales included. Thus, in comparison to simple pairwise correlations, the chain graph model allows us to assess correlations between variables while controlling for others, along with handling causal associations between blocks of variables and non-causal associations within blocks.

Figure 1 shows the underlying block-recursive structure for the analysis, with arrows pointing from one block to blocks occurring later in time, thereby giving direction (i.e., timewise causality) to any associations between variables from different blocks. Associations between variables within a block are without direction. The PSEQ subscales and the selected background variables were placed in the various blocks based solely on timewise order of the variables, from right to left. Block A consisted of gender (female, male). Block B consisted of information on students' previous experience: Admission track (grade-based, other qualifications), and Previous teaching experience (no, yes). Block C consisted only of

students' choice of Major teaching subject (Danish, mathematics, English). Finally, Block D consisted of the seven practice self-efficacy subscales.

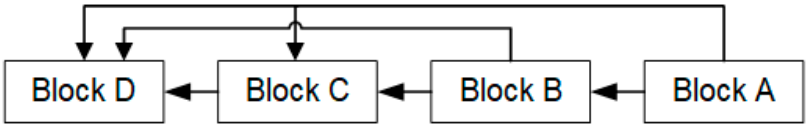


Figure 1. The block structure of the chain graph model, with arrows illustrating the time-wise causal structure for the analysis. Notes. Block D represents the most recent data (the PSE subscales), which can we affected by data in block C, B and A occurring prior to the PSE measurement.

The DIGRAM software package version 5.05.07 was used to define and test the chain graph model (Kreiner, 2003, 2014). The correlational structure of the model was determined based on statistically significant correlations using partial Goodman-Kruskal gamma (γ) correlations (Partial correlations supplies correlations between two variables while controlling for the effects of other variables), as these are appropriate for ordinal categorical data (Davis, 1967; Goodman & Wallis, 1954; Kreiner, 1987). γ coefficients above 0.30 can be regarded as a strong association, and γ coefficients below 0.15 a weak association (Kreiner, 2007), though there are no set limits. To define a simpler starting model than the full block-recursive model, we used an automated screening procedure for high-dimensional contingency tables (Kreiner, 1986). Next, we employed a stepwise manual model selection strategy for the purpose of improving the starting model towards the end goal of identifying an adequate model for the data (Kreiner, 1986). The strategy included both backwards and forwards model selection. Backwards model selection tests the relationships between all variables in the model for conditional independence, and if two variables were conditionally independent given the other variables in the model, the edge between these variables would be deleted. Forwards model selection tests the relationships between all variables in the model for conditional independence, and if two variables where conditionally dependent given the other variables in the model, the edge between these variables would be added. At each step the model would thus change and all relationships would be tested anew. From the model resulting from the automated screening procedure, we conducted backwards model selection until no further edges could be deleted. This was followed by forwards model selection until no more edges could be added, and so on until no more edges could be added or deleted. Decisions about including or deleting edges in the manual search strategy were based on both the strength of the evidence (i.e., p-values), the strength of the associations for ordinal variables (i.e., γ coefficients), and subject matter knowledge for relevance of associations. Thus, weak associations (i.e., $\gamma < 0.15$) were not considered unless the evidence was strong, while strong associations (i.e., $\gamma > 0.30$) were not considered if the evidence was weak. The strength of the evidence was evaluated on a continuum distinguishing between weak ($p < 0.05$), moderate ($p < 0.01$) and strong ($p < 0.001$) evidence, as recommended by Cox et al. (1977).

The final model was confirmed by testing the necessity of all the included associations as well as testing the adequacy of the included associations. Lastly, partial γ-correlations were estimated for all associations in the final chain graph model.

The problem of estimating the γ-coefficients and p-values with asymptotic methods was resolved by using a Monte Carlo procedure with 1000 samples to obtain exact p-values. Multiple testing was dealt with by controlling the false discovery rate (FDR) with the Benjamini–Hochberg procedure (Benjamini & Hochberg, 1995).

3. Results

The final chain graph model showing associations and the lack thereof (i.e., conditional dependence/independence) is shown in Figure 2 including the partial γ-correlations. No causality, other than that imposed by time, is implied in the graph (from right to left; cf. the recursive structure in Figure 2).

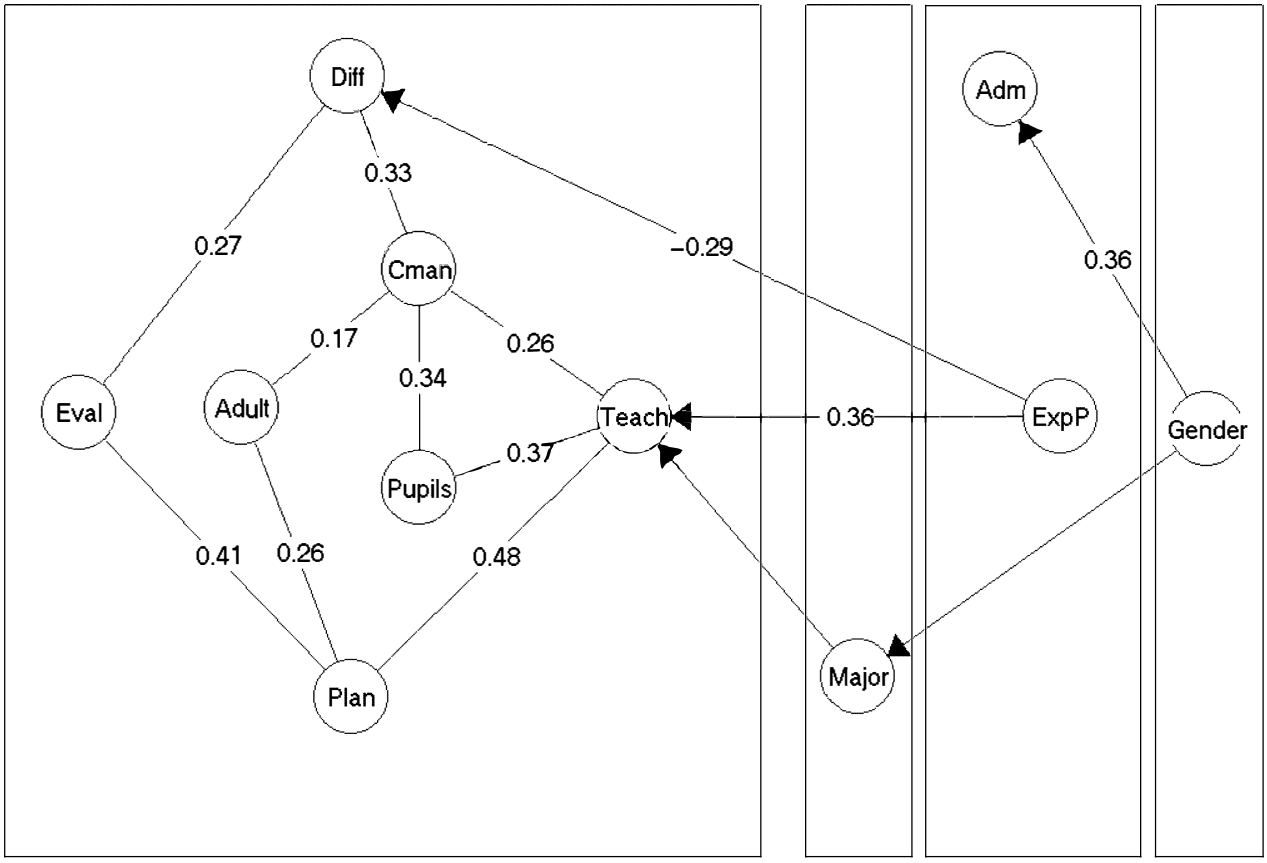


Figure 2. The final chain graph model. *Notes.* All edges and arrows represent significant conditional associations between variables given the remaining variables (conditional dependence), e.g., *Planning and preparation PSE* is conditionally associated with *Teaching in itself PSE* given the remaining variables in the model. The absence of a line or an arrow in the model shows that the two un-connected variables are not associated given the remaining variable (conditional independence), e.g., *Adult collaborators PSE* and *Pupils PSE* are conditionally independent when taking into account the remaining variables in the model. Correlations are partial Goodman-Kruskal gamma-coefficients (γ).

3.1. Associations Between PSE Scores

As expected, we found only positive associations between the PSE subscales (Figure 2). However, the associations between specific PSE scores did not match our a priori expectations in all cases. Thus, *Planning and preparation* PSE and *Evaluation and development* PSE were strongly and positively associated (γ = 0.41). Also, *Class management* PSE and the *Teaching in itself* PSE were also positively, but less strongly, associated (γ = 0.26), and both were strongly associated with *Pupils* PSE (γ = 0.34 and γ = 0.37, respectively) (H3). However, the expected association between *Planning and preparation* PSE and *Differentiation* PSE, was not found in the final model.

Additional positive associations between the seven PSE subscales were present in the final model (Figure 2). These were of varying strength; from a weak association between *Class management* PSE and *Adult collaborators* PSE (γ = 0.17) to a very strong association between *Planning and preparation* PSE and *Teaching in itself* PSE (γ = 0.48).

3.2. Conditional Dependence/Independence of PSE Scores on Background Variables

None of the seven PSE subscales were conditionally associated with gender or admission track.

Only the *Teaching it itself* PSE and the *Differentiation* PSE subscales were conditionally dependent on background variables, while the remaining five PSE subscales were conditionally independent of background variables given *Teaching in itself* PSE and/or the *Differentiation* PSE and, in some cases additional PSE subscales (see Tables S1 and S2 in the Supplementary Materials for variables separating each PSE subscale from background variables).

Differentiation PSE was conditionally dependent on teaching experience prior to teacher education given *Teaching in itself* PSE. This direct effect of prior teaching experience on *Differentiation* PSE was medium strong and negative ($\gamma = -0.29$, exact $p < 0.001$, 99% CI 0.000–0.007), meaning that prior teaching experience was associated with lower PSE for *Differentiation* than was a lack of prior teaching experience.

Teaching in itself PSE was conditionally dependent on teaching experience prior to teacher education given *Differentiation* PSE and Major ($\gamma = 0.36$, exact $p < 0.001$, 99% CI 0.000–0.007). Thus, while the direct association of prior teaching experience on *Teaching in itself* PSE was strong and positive ($\gamma = 0.36$), meaning that prior teaching experience was associated with higher PSE for *Teaching in itself* than was a lack of teaching experience, this should be examined further in order to uncover whether this association was global or whether there were local associations in strata defined by *Differentiation* PSE or Major (We could have chosen to describe the conditional dependence of Teaching in itself PSE on Major given teaching experience prior to teacher education instead. However, as we had no prior expectations with regard to associations between Major and any of the PSE subscales, we chose to describe further the conditional association between prior teaching experience and Teaching in itself PSE). Stratification by *Differentiation* PSE and Major showed evidence of a higher order interaction resulting in local associations of prior teaching experience and *Teaching in itself* PSE in strata defined by the conditioning variable Major (Table 3). Thus, the association between prior teaching experience and *Teaching in itself* PSE was only moderate in strength for student teachers' who had chosen Danish as their Major teaching subject ($\gamma = 0.26$), strong for students who had chosen mathematics ($\gamma = 0.55$), and very strong for students who had chosen English ($\gamma = 0.84$)

Table 3. Local test results for association between prior teaching experience and teaching in itself PSE for strata defined by Major.

Major	***df***	γ	***p* Exact**
Danish	60	0.26	0.0050
Mathematics	34	0.55	<0.0001
English	29	0.84	<0.0001

A further stepwise analysis of pairwise collapsibility showed that the three local associations between prior teaching experience and *Teaching in itself PSE* defined by Major could be reduced to two local conditional associations (Table 4): A strong and positive association ($\gamma = 0.38$, SE 0.077) for the combined group of students who had chosen Danish or mathematics as their Major teaching subject, and a very strong association ($\gamma = 0.84$, SE 0.084) for the group of students who had chosen English as their Major.

Table 4. Collapsibility of the three local associations between prior teaching experience and teaching in itself PSE in strata defined by Major.

Major Groups Tested	Critical *p*	*p*	Decision
Danish and Mathematics	0.0167	0.0667	Collapsed
Danish + Mathematics and English	0.0250	0.0001	Collapse not possible

Notes. The stepwise analysis collapses the categories where the *p*-value is larger than the critical *p*-value controlling the FDR at 0.05 at each step (Kreiner, 2014). First, three tests are conducted comparing the three nominal categories (Danish & math, Danish & English, and math & English). Second, Danish and mathematics are then collapsed into a single category, because the comparison of these categories provided the largest *p*-value and it was higher than the critical *p*-value. Third, the collapsed category of Danish and mathematics is compared to English, which end the analysis, as they cannot be collapsed (i.e., the *p*-value is well below the critical *p*-value).

4. Discussion

The study investigated the relationships between practice self-efficacy dimensions among student teachers. While earlier studies have explored how teacher self-efficacy dimensions are related (Poulou, 2007; Klassen & Chiu, 2011; Burgueño et al., 2019; Khairani & Makara, 2020; Berg et al., 2023), they have not used student-tailored self-efficacy instruments aimed at the field practice elements of teacher education.

We found that many, but, importantly, not all, practice self-efficacy dimensions were directly related, when simultaneously analyzing all dimensions together with the background factors in a graphical model setting. The strongest associations were found among PSE dimensions related to the core teaching-related domain of practice self-efficacy, such as between *Planning and Preparation PSE* and *Teaching in itself PSE,* as well as between *Class management PSE* and *Differentiation PSE*. Associations within this domain also appear more widespread, although not all dimensions within were directly related. Notably, two PSE dimensions stand out—*Planning and Preparation PSE* and *Class Management PSE*—as these were directly associated with various other self-efficacy dimensions—both within and across domains. In contrast, relational PSE dimensions, such as *Adult collaborators PSE*, and *Evaluation and development PSE* were more peripheral, as these dimensions showed either weaker or fewer associations with other PSE dimensions.

The results have several important implications for the field of teacher education. When considering results collectively, the positive associations between some of the practice self-efficacy dimensions and the differences in strength of these associations point to areas where teacher education may most effectively intervene to enhance practice self-efficacy before and during field practice. For example, *Planning and preparation PSE* was strongly associated with both *Teaching in itself PSE and Evaluation and development PSE*, and moderately associated with *Adult collaborators PSE.* Thus, by targeting and focusing on providing student teachers opportunities for positive vicarious experiences, emotionally and physiologically positive experiences (i.e., non-stressful and/or eliciting positive emotional states), or even mastery experiences, in relation to planning and preparational activities, teacher education programs may enhance directly their practice self-efficacy for *Planning and preparation* as well as their practice self-efficacy for *Teaching in it-self*, *Evaluation and development*, and collaboration with colleagues and parents (*Adult collaborators).* Actually, a focus on enhancing practice self-efficacy for planning and preparation (and teaching in itself) is also in line with the national learning objectives for first-year field practice in the Danish teacher education (UCL Erhvervsakademi og Professionhøjskole, 2024). These emphasize that student teachers should be able to identify and gather relevant knowledge on teachers' planning and teaching, and one mean towards this end, which could also enhance their self-efficacy, could be vicarious experiences of planning and teaching both before and during field practice in the form of having access to particularly good examples of teaching plans, and to observe planning activities and the resulting teaching activities.

Thus, teaching plans and observation would be representations and decompositions of practice to be analyzed and discussed (Cohen & Berlin, 2020). The learning objectives also emphasize that student teachers should be able to plan, implement and evaluate short teaching sequences. To enhance self-efficacy, and thus also maximize the potential for successful learning outcomes and achievement of the learning objectives teacher education, on-campus preparation for field practice might include possibilities to enact core skills and practices related to planning and preparation (Klette et al., 2017; Schutz et al., 2018). This could for example be in the form of role play, teaching of fellow student teachers to train various micro activities or virtual reality enhanced practice session (Howell & Mikeska, 2021; Klette et al., 2017; Schutz et al., 2018). Such learning activities provide student teachers with opportunities to practice activities approximating real teaching activities, yet in a controlled and artificial setting where a sense of mastery can be planned" by educators by degree of realism and degree of difficulty. Furthermore, the setting would be a relatively training and learning-wise safe environment to practice in, as all the possible unexpected situations and disturbances occurring in the real school context would be absent. Thus, the chance of successful and emotionally positive sessions contributing gradually towards an enhanced sense of self-efficacy is maximized. It follows naturally that a prudent line of future research would be longitudinal studies investigating directional influences among PSE domains and dimensions.

Obviously, the above type of activities is not easily achieved in the school field practice context, as this is a complex and unpredictable context where unpredictable situations and issues arise. Thus, the more important it is to nurture positive experiences in the early stages of teacher education, and particularly in the first year, so that the chance of the field practice experience turning into a negative practice encounter experience (i.e., practice shock; Haugen & Hestbek, 2015) and at the same time a negative effect on self-efficacy is minimized (Bandura, 1997). This could be further enhanced by supervision and guidance, for example, by breaking down teaching sessions of both the teacher and the student teacher and discussing parts of them: What did you/I want to achieve? What happened? What contributed to this? What can you/I do differently to change the outcome? This would both train teacher practices, how to cope in unexpected situations and enhance students' reflective capacity. As shown by Ronfeldt et al. (2018), the ecologically valid environment that makes up the field practice is not in itself enough to ensure the best learning outcome with regard to practical skills and development of teacher practices. They suggest that role models—or in Bandurian terms positive vicarious experiences, also play a role. Likewise, interventions targeted to enhance *Class management PSE* may also have some effect on the student teachers' practice self-efficacy in relation to *Differentiation*, collaboration with colleagues and parents (*Adult collaborators)*, the relational engagement with *Pupils*, and Teaching in it-self, as these four PSE dimensions are all associated with *Class management PSE* to varying degrees. Experiments that test the effects of specific interventions designed to enhance a specific dimension of practice self-efficacy, such as concrete activities or forms of learning, are warranted. Such experiments are needed to learn about the mechanisms of enhancement and whether some dimensions are malleable indirectly through the targeting of other dimensions.

In addition, we suggest future research should focus on the effect of practice self-efficacy, for instance exploring individual level changes, on learning outcomes related to practical core skills of teachers over time. Such research would show whether there is the same strong association between practice self-efficacy and outcome with regard to the practical core skills in teacher education, as previous research has documented between academic self-efficacy and academic learning outcomes in many areas of higher education (e.g., Richardson et al., 2012).

The negative association between prior teaching experience and *Differentiation PSE* is interesting, as we did not expect any negative associations between PSE dimensions and prior teaching experience based on previous research. However, it may simply be that prior teaching experience has shown the student teachers that differentiation is a difficult task requiring both didactical and subject-wise knowledge and skills as well as knowledge of the pupils' proficiency in a subject, and thus their beliefs in their own abilities are lowered compared to the beliefs of the student teachers without such experience. The experience in schools thus serves as a source of adjustment of practice self-efficacy in relation to differentiation, which is in accordance with self-efficacy theory (Bandura, 1997). This is further supported by the fact that we have only included first-year students prior to any field practice as part of the teacher education, and the difference in practice self-efficacy for differentiation therefore more than likely refers back to their prior teaching experience in school. Currently, our explanation cannot be supported by previous research, since it is as a new finding. Therefore, we suggest that future research seeks to corroborate this finding. A first step would be to replicate the study with new samples of first-year students. It could also be substantiated through an in-depth qualitative study on the relationship between prior teaching experience and Differentiation PSE. Lastly, future longitudinal and prospective studies focusing on the relationship between experience and *Differentiation PSE* (e.g., teaching experience prior to teacher education -> *Differentiation PSE* prior to field practice in year 1 -> field practice and other teacher education experiences -> *Differentiation PSE* prior to field practice in year 2 -> and so on) would further elucidate how this relationship potentially evolves.

The positive local associations between prior teaching experience and *Teaching in itself PSE* given the students teachers' chosen Major are interesting findings in two ways. Firstly, because the inclusion of all seven practice efficacy dimensions in the multivariate model meant refined the results from Nielsen et al. (2024b), as this resulted in only *Teaching in itself PSE* being directly positively associated with prior teaching experience, while the remaining PSE dimensions—excluding *Differentiation*, which was negatively associated (c.f. the above)—were all indirectly associated with prior teaching experience through *Teaching in itself PSE*. Secondly, while the positive association is not in itself surprising as we would expect that student teachers who had very negative experiences with teaching prior to entering higher education would not apply for admission to teacher education. Thus, the finding is as expected theoretically: that student teachers without teaching experience will be less self-efficacious with regard to teaching than student teachers with (to some degree) positive experiences (Bandura, 1997).

The finding that the association between prior teaching experience and PSE for *Teaching in itself* was substantially stronger for student teachers who had chosen English as their major teaching subject, compared to the students that had chosen Danish or mathematics, was unexpected. It may be that some or even many student teachers who choose English as their major in reality reject Danish and mathematics, and then select the only remaining major. Thus, choosing English is not a straightforward choice, but a result of a somewhat more complex process. This appears a credible explanation, as there are only three choices of major, and, in general, the Danish population is proficient in English. They may, through their prior teaching experience as, e.g., a substitute teacher, have experienced that their English proficiency level combined with the also rather good level of English among many pupils, made it easier to teach the subject of English compared to other subjects such as Danish or mathematics. This explanation is supported by the common, but often unstated, popular notion that it is easy as a substitute teacher to come up with teaching materials and opportunities in the form of films, texts and song, which can then be discussed in English with the pupils. In the subjects of Danish and mathematics, there is an equally common

notion that these subjects require more specialized knowledge of linguistic elements or rules of mathematics, in order to be able to teach them. This difference may be based on naïve notions about teaching English stemming from a functional and communicative view of language teaching, which focuses less on accuracy and correctness and more on making oneself understood, for the emerging teacher. While we cannot point to any implications for the field of teacher education for this result, an area for future research would be a qualitative exploration of first-year student teachers' understandings of their sense of English/Danish/mathematics proficiency, what it requires to be able to teach English/Danish/mathematics, and how this connects to their sense of *teaching in itself PSE* through previous teaching experience, in comparison across the three majors.

Limitations

The study, while the first of its kind, does have limitations, or what appears to be limitations. One limitation is related to the small N in the sample. It is obvious that a larger N would be preferable due to issues of power. However, as we use Monte Carlo simulation to obtain exact *p*-values, this to some degree mitigates this issue. A second limitation relates to the scope of the study, as data was only collected in a single national context. Thus, we cannot claim that the results bear relevance to teacher education in other countries, as the educational culture and the nature of teacher education differs across countries. Cultural and structural educational differences in other countries limit generalizability. However, this is the case for many constructs investigated within education and not isolated to the current study (c.f. (Nielsen et al., 2021) on critical thinking, and (Nielsen et al., 2025) for statistical anxiety). A third, seemingly, limitation relates to our model design, where the seven PSE dimensions were placed at the same recursive level due to the time-wise causality implied in relation to the background variables. In the context of teacher education, it may be that the seven PSE dimensions are connected in a more complex recursive structure. For example, if practice self-efficacy in relation to what might be considered foundational skills, such as planning or class management, was to precede more self-efficacy in more complex skills and competencies such as differentiation, we cannot investigate these issues with the current design. To investigate this, data with temporal variation are necessary to tap into questions of sequential or reciprocal relationships between PSE dimensions. A fourth possible limitation also relates to our model design and the time-wise causality implied in the model. Thus, while it is factual that when considered in a timeframe, gender was determined prior to previous teaching experience and admission track for teacher education, while the latter two were determined prior to students choosing their major, which also occurred prior to the measurement of PSE, this causal order represents the occurrence order within the "lives" of the student teachers. However, all data entered into the model were collected through the same survey. This may, by some, be considered a limitation of the study. We do not find it to be a huge limitation, as it is not an issue that students would forget this information in the short time span of approximately five weeks from the time of admission notice to the survey.

5. Conclusions

Previous studies suggest that among student teachers all self-efficacy dimensions are intricately related. Our findings provide evidence that many, but not all practice self-efficacy dimensions are associated among student teachers before their first field practice, with the stronger associations found among dimensions within the teaching-related domain. In particular, we found that two dimensions—*Planning and preparation PSE* and *Class management PSE*—were related to numerous other self-efficacy dimensions—also some from other domains. Taken together, these findings pinpoint areas that teacher education

could focus on with targeted activities to effectively improve practice self-efficacy before field practice, thereby limiting the risk of a negative practice encounter experience. In addition, we found that all PSE dimensions were independent of background characteristics (i.e., gender, admission track, major, and prior experience), except *Differentiation PSE* and *Teaching in itself PSE*, which both were related to prior teaching experience. Specifically, we found that prior experience was positively associated with *Teaching in itself PSE* and negatively associated with *Differentiation PSE*.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/psycholint7030059/s1, Table S1. Testing the necessity of the associations in the final chain graph model. Table S2. Testing the adequacy of the associations in the final chain graph model.

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Institutional Review Board Statement: No ethical approval was needed for research involving only survey data in the Danish context. Participating student teachers were informed of their right to withdraw from the study at any time prior to data anonymization, as well as of all other rights and of how their data would be treated in accordance with current European data protection regulations.

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Data Availability Statement: The dataset presented in this article is not readily available because the data are part of an ongoing study. Requests to access the datasets should be directed to the corresponding author.

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